> **Important**

> **Warning**

[[]]

```
```

API

🤦‍♂️ this section will discuss about Web APIs

- stands for Application Programming Interface
- set of rules that allows application to communicate with the server
- API defines how HTTP (Hypertext Transfer Protocol) request messages are send and how responses are returned in the form of JSON (JavaScript Object Notation) format HTTP request methods GET, POST, PUT, DELETE
- previous protocol is SOAP (Simple Object Access Protocol) and SOA (Service-Oriented Architecture), now is a more direct protocol REST (Representational State Transfer)
- by API data can be created in one place and shared to multiple locations on web

Building API

HTTP

HTTP is the protocol for communicating between client (browser) and server (place you access the data in), resources for communicating can be almost everything.

client opens the connection while the server closes the connection
http_request_response_cycle.png

Format for Request and Response Messages

```
Status Line = Initial Request Line
```

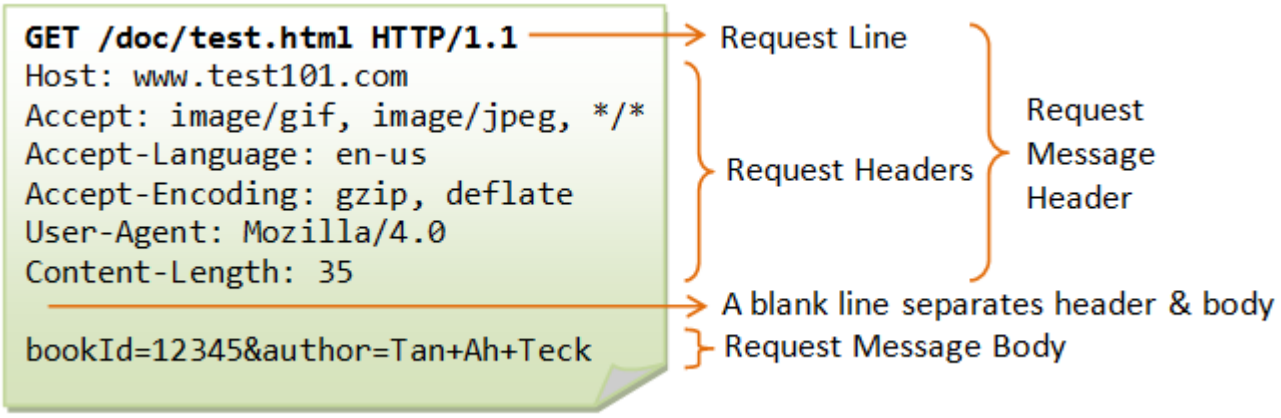
Header Fields

provide information about the request or response, provide information about the body as well, the MIME-type (text/html, application/json, text/plain, text/css, image/gif) and the content length, the number of bytes

Message body

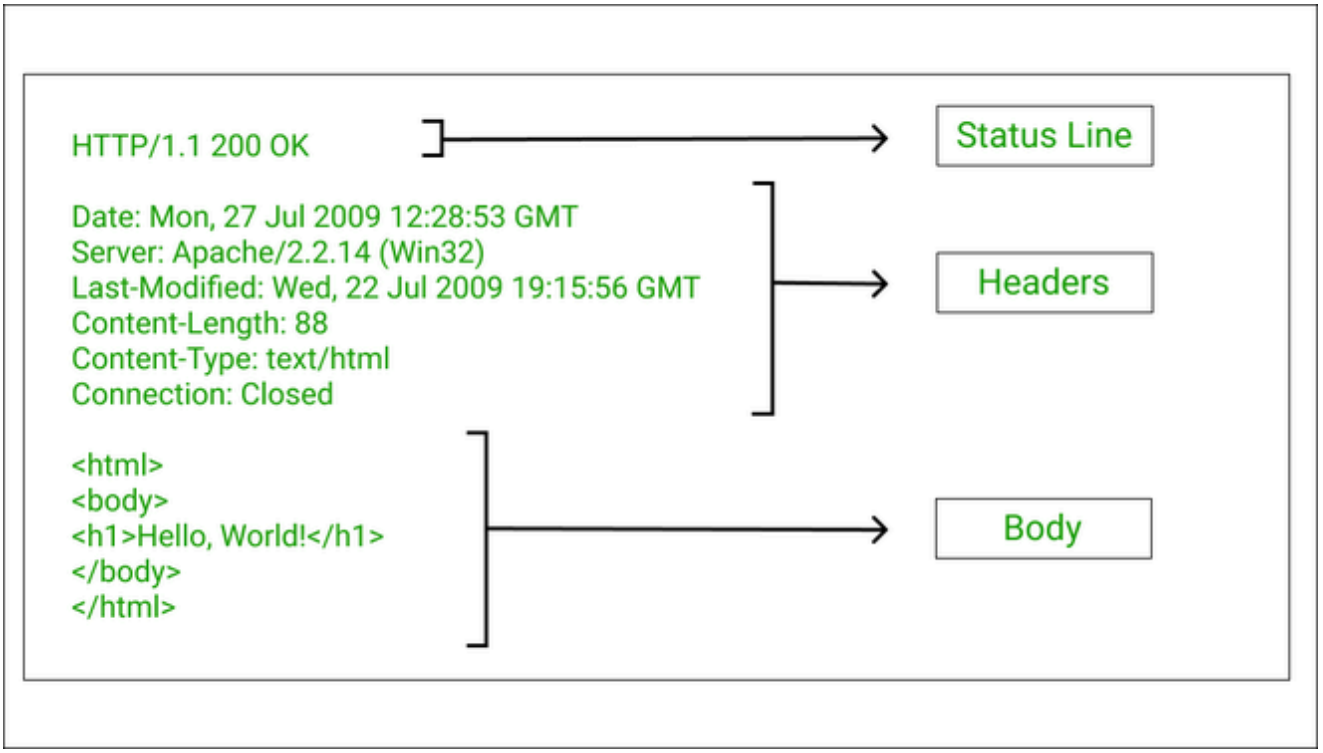
is where the requested resource is returned, if failed, there will be explanatory text for an error occurred.

Request Message



- method name(GET, POST, HEAD)
- path of the requested resource,
- the version of HTTP being used. eg GET / HTTP/1.1

Response Message



- HTTP version
- Response status code that gives the result of the request, and a reason which describes the status code.

Status Line = Initial Request Line

Request Methods

1. GET, to retrieve information/data from given server, no effect on the data
2. POST, create and send data to the server
3. PUT, replace all current representations of target resource, modify or update data
4. DELETE, removes data